

Complete Analytical Framework

All statistical models used in hybrid dysfunction analysis

Model_ID	Model_Type	Model_Formula	Purpose
RF-1	Random Forest ¹	Weight_loss ~ 19 immune genes	Train immune-health model
RF-2	Random Forest ¹	Weight_loss ~ 19 immune genes (train-test)	Validate predictive accuracy
RF-3	Random Forest ¹	Weight_loss ~ 19 immune genes (cross-validation)	Bootstrap validation
RF-4	Random Forest ¹	Weight_loss ~ 19 immune genes (final model)	Final model for field application
FIELD-1	Random Forest	Predicted_WL ~ 19 immune genes (field application)	Apply to wild mice (n=335)
FIELD-2	Random Forest	Model validation (infection status, species, intensity)	Validate field predictions
PL-1	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Sex (Complete dataset)	Test sex-specific hybrid effects
PL-2	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Sex (Uninfected only)	Test constitutive hybrid costs
PL-3	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Infection (Infection dominance)	Test infection vs genetic effects
PL-4	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Sex (Infected only)	Test infection-specific hybrid costs
PL-5	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Sex (Males only)	Confirm male-specific effects
PL-6	parasiteLoad ²	response ~ HI + hHe + $\alpha \times hHe$ Sex (Females only)	Confirm female protection

¹ Random Forest models from Webster et al. (in review)

² parasiteLoad package implements Baird et al. (2012) framework

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Model_ID	Model_Type	Model_Formula	Purpose
LM-1	Linear Regression	Predicted_WL ~ HI × hHe	Test genetic baseline effects
LM-2	Linear Regression	Predicted_WL ~ Sex + HI + hHe + Infection	Test main effects model
LM-3	Linear Regression	Predicted_WL ~ Sex × HI + Sex × hHe + HI × hHe + Sex × Infection + HI × Infection + hHe × Infection	Test interaction effects (best model)
LM-4	Linear Regression	Predicted_WL ~ Sex × HI × hHe × Infection (full model)	Test full interaction model
LM-5	Linear Regression	Predicted_WL ~ Sex × (HI + hHe) × Infection + HI:hHe (focused)	Test biologically focused model

¹ Random Forest models from Webster et al. (in review)

² parasiteLoad package implements Baird et al. (2012) framework